ABOUT US

AutoMind Dynamics is a deep-tech startup incubated at **ART-PARK, Indian Institute of Science (IISc), Bengaluru,** building next-generation autonomous vehicles and intelligent software for real-world deployment. We design high-performance autonomy solutions for both indoor and outdoor environments by tightly integrating AI-driven software with robust hardware and embedded systems. Our mission is to embed intelligence into mobility—delivering autonomous systems that are practical, safe, and scalable, enabling industries to operate smarter, faster, and more efficiently.

OUR PRODUCT

- Autonomous ground robot—ready for Research and academia
- Rapid deployment with minimal setup and configuration
- Built-in autonomy stack covering perception, planning, navigation, control, and obstacle avoidance
- Engineered for both research and real-world experimentation, powered by open-source ROS2 frameworks
- Modular payload support, enabling easy integration of custom sensors and research hardware

WE OFFER

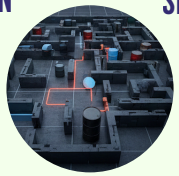
SLAM

AUTONOMOUS DRIVING

SENSOR FUSION

SIMULATION PLUGIN

AI BASED OBJECT DETECTION


PATH PLANNING

ARTPARK
AI & Robotics Technology Park @ IISc

info@automindynamics.com
www.automindynamics.com

SPECIFICATIONS

DIMENSIONS	390 x 348 x 190 mm (15.35 x 13.7 x 7.4 in)	**RUN TIME**	Performance: 4 hrs Basic Usage: 6 hrs
EXTERNAL POWER INTERFACE	12V 5A	**ODOMETRY ACCURACY**	95-97%
MAX. PAYLOAD	5 kg (11 lbs)	**BATTERY**	3S 15 Ah
MAX SPEED	1.36 m/s (4.9 Km/h)	**SENSORS**	Stereo Camera, IMU, LIDAR(2D,3D)

COMMUNICATION	Ethernet, USB 3.0
FRAMEWORK	Packaged with ROS2
COMPUTE UNIT	Jetson Orin Super, Orin Nx 8/16
FEEDBACK	Battery, Motor Current, Wheel Velocity & IMU

PLUGINS	MATLAB, ROS 2 (Humble / Jazzy), Isaac Sim, PyBullet, Webots, Gazebo
COMMUNICATION INTERFACES	Wi-Fi 5 & BLE USB-A (External) & USB-C (Debugging) RJ45 Ethernet
DISPLAY INTERFACE	HDMI
POWER INTERFACE	LP-16 Connectors

SENSOR ECOSYSTEM

Depth & AI Cameras: Full support for OAK-D, Intel RealSense, ZED, and YDLIDAR series for advanced spatial AI and stereo vision.

LiDAR (2D/3D/4D): Pre-configured drivers for Livox (Mid/Avia/HAP), Velodyne, Ouster, and YDLIDAR for high-fidelity mapping.

RealSense · LIVOX · Velodyne Lidar · OUSTER · YDLIDAR

Precision Positioning: Ready for RTK-GNSS, Neo-M8N, and multi-constellation GPS modules.

Radar: Integrated support for mmWave and ARS series long-range detection.

READY FOR RESEARCH

FOR STUDENTS:
- Learn real-world robotics stacks
- Implement SLAM, perception & control
- ROS 2 based hands-on deployment

FOR RESEARCHERS:
- Skip hardware bring-up
- Prototype perception & navigation fast
- Open-source ROS 2 foundation

TOP VIEW

SIDE VIEW

FRONT VIEW

SAFETY

Integrated BMS (Battery Management System): Continuous health checks for Cell Balancing, State of Charge (SOC), and thermal limits.

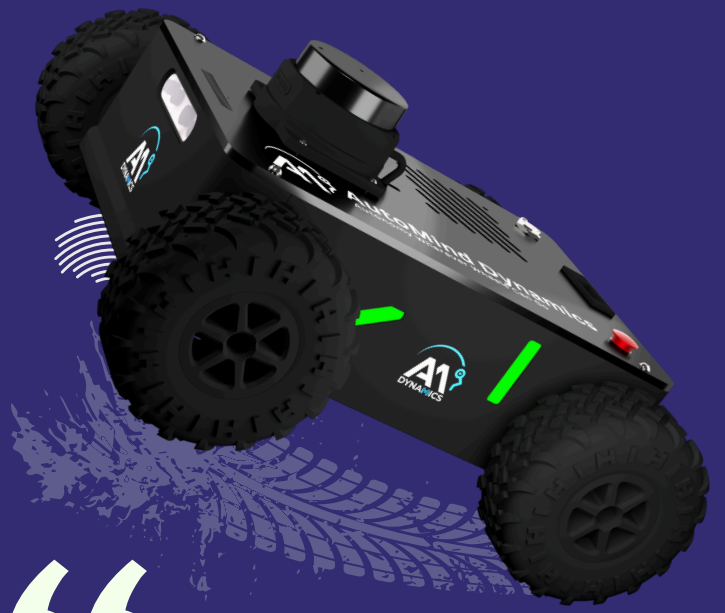
Smart Charging Protection: Integrated over-voltage and over-current safeguards ensure battery health and safe indoor charging.

Emergency Kill Switch: Physical E-Stop for immediate hardware-level power cutoff.

Multi-State LED Indicators: High-visibility status LEDs provide instant visual feedback on system health, battery levels, and ROS 2 connectivity.

GEFIER R1

EXPERIENCE THE FUTURE OF AUTONOMY



INDOOR MOBILE ROBOT

From Equation to Execution—Instantly

Bypass the build, deploy your custom controls, and validate your results on day one.